

31. What is a Function in Python?

Answer:

A **function** is a reusable block of code that performs a specific task. Instead of writing the same code multiple times, you can write it once inside a function and call it whenever needed.

Advantages:

- Reduces code duplication.
- Makes programs easier to read and maintain.
- Improves code reusability.

Syntax:

```
def function_name():
```

```
    # code
```

Example:

```
def greet():
```

```
    print("Welcome to Python!")
```

```
greet()
```

Output:

Welcome to Python!

Interview Tip: Functions are created using the **def** keyword.

32. What is the difference between Parameters and Arguments?

Answer:

- **Parameters** are variables defined in the function definition.
- **Arguments** are the actual values passed when calling the function.

Example:

```
def add(a, b): # a and b are parameters
```

```
    return a + b
```

```
print(add(10, 20)) # 10 and 20 are arguments
```

Output:

30

Interview Tip: This is one of the most commonly asked Python interview questions.

33. What is the difference between return and print()?

Answer:

return

print()

Sends a value back to the caller Displays output on the screen

Can be used in further calculations Cannot be reused directly

Ends the function Does not end the function

Example:

```
def square(x):
```

```
    return x * x
```

```
result = square(5)
```

```
print(result)
```

Output:

25

Interview Tip: Use return when another part of your program needs the result.

34. What are Default Arguments?

Answer:

A **default argument** is a parameter with a predefined value. If no argument is passed, Python uses the default value.

Example:

```
def greet(name="Guest"):
```

```
    print("Hello", name)
```

```
greet()
```

```
greet("Bhagya")
```

Output:

Hello Guest

Hello Bhagya

Interview Tip: Default arguments make functions more flexible.

35. What are `*args` and `**kwargs`?

Answer:

- **`*args`** allows a function to accept any number of positional arguments.
- **`**kwargs`** allows a function to accept any number of keyword arguments.

Example:

```
def total(*numbers):  
    print(sum(numbers))
```

```
total(10, 20, 30)
```

Output:

60

Example using `**kwargs`:

```
def student(**details):  
    print(details)
```

```
student(name="Bhagya", age=20)
```

Output:

```
{'name': 'Bhagya', 'age': 20}
```

Interview Tip: `*args` and `**kwargs` are frequently asked in Python interviews because they make functions more flexible.

